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(34)

CO2 -AT1

Slot 'D'

## SIMATS ENGINEERING

### ASSESSMENT TOOL 1: Analytical Problem Solving

**COURSE NAME:** Cloud Computing and  
Big Data Analytics

**COURSE CODE:** CSA1510

#### COURSE OUTCOME COVERED

**CO2:** Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)  
Question Count: 5

**Total Marks: 50**

#### Analytical Problem Solving Rubric

| Criteria                 | Weightage | Level 4 - Exemplary                                             | Level 3 - Proficient                   | Level 2 - Developing            | Level 1 - Beginning         |
|--------------------------|-----------|-----------------------------------------------------------------|----------------------------------------|---------------------------------|-----------------------------|
| Problem Interpretation   | 20%       | Interprets all technical requirements accurately                | Interprets most requirements correctly | Partial interpretation          | Misinterprets the problem   |
| Analytical Reasoning     | 25%       | Strong reasoning with clear cause-effect analysis               | Good reasoning with minor gaps         | Basic analysis only             | Weak or no analysis         |
| Method Selection         | 20%       | Chooses highly appropriate virtualization and security concepts | Chooses mostly suitable concepts       | Partially suitable concepts     | Inappropriate approach      |
| Solution Quality         | 20%       | Provides feasible and technically sound solution                | Reasonable solution with minor issues  | Basic solution with limitations | Unclear or invalid solution |
| Presentation of Analysis | 15%       | Well-structured and technically precise                         | Generally clear                        | Somewhat unclear                | Difficult to follow         |

**Total Marks Awarded:**

#### Code of Conduct

I Abdul Razak Shaik (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Abdul Razak Shaik

①. Compare Type-1 and Type-2 Hypervisors for a cloud Datacenter Hosting Mission - Critical Applications.

A.

Feature:-

Installation:-

Type 1:- Runs directly on hardware

Type 2:- Runs on top of a host OS

Performance:-

Type 1:- High, low overhead

Type 2:- Lower due to host OS overhead.

Security:-

Type 1:- More secure, smaller attack surface

Type 2:- Less secure because host OS can be attacked.

Manageability:-

Type 1:- Supports centralized management and live migration.

Type 2:- Suitable for personal / testing Environment.

Examples:-

Type 1:- VMware ESXi, Microsoft Hyper-V

Type 2:- Oracle VM Virtual Box, VMware Workstation.

## Analysis:-

1) Performance :- Type-1 provides near-native performance.

2) Security :- Better isolation and fewer vulnerabilities.

3) Manageability :- Supports clustering, live migration and centralized monitoring.

## Justification:-

→ For mission-critical cloud applications, Type-1 hypervisor is the better choice because it provides higher performance, stronger security, and Enterprise Level Management features.

## ② Workload isolation using Virtualization Architecture :-

Virtualization allows multiple virtual machines (VMs) to run independently on the same physical server.

### How isolation is achieved:-

① Each VM has its own virtual CPU, memory, storage and network.

② The hypervisor prevents one VM from directly accessing another VM's resources.

virtual networking allow, separate VLANs and security groups for finance, healthcare, and student portals.

Two major attack surfaces:

1. Hypervisor attacks - An attacker exploits vulnerabilities in the hypervisor to gain access to multiple VMs
2. VM Escape attacks:- Malicious code escapes from one VM to the host or another VM.

Conclusion:-

Virtualization provides strong isolation, but securing the hypervisor and virtual networks is essential.

③ VM Host Utilization Analysis:-

Given:-

- CPU = 92%
- Memory = 88%
- Storage I/O latency = 35ms
- VM boot time = 170s

Likely bottlenecks:-

- CPU (92%) - processor is overloaded, causing slow execution.
- Memory (88%) - high memory usage may cause swapping
- Storage Latency (35ms) - disk access is slow, delaying VM startup.

• VM boot time (170s) - indicates storage and CPU bottlenecks

### Recommended corrective actions:-

- Add more CPU resources on distributive workloads.
- Increase RAM or reduce memory overcommitment.
- Upgrade to SSD/NVMe storage
- Balance workloads across additional hosts.
- Enable resource monitoring and auto scaling.

### Conclusion:-

The primary bottlenecks are CPU overload and high storage latency, leading to slow VM performance and long boot times.

### ④ Software Defined Datacenter (SDDC) vs Traditional Datacenter.

| Traditional Datacenter | Software Defined Datacenter. |
|------------------------|------------------------------|
| Manual Configuration   | Automated provisioning       |
| Hardware dependent     | Software Controlled          |
| Slow deployment        | Rapid deployment             |
| Limited scalability    | Highly Scalable.             |

## Virtualization improvement:-

### compute Virtualization

- created vms quickly
- Better resource utilization

### Storage Virtualization:-

- pools multiple storage devices
- Easier Backup and Expansion.

### Network Virtualization

- creates virtual network instantly
- Support Software - defined networking (SDN)
- faster security policy deployment.

### Analysis:-

SDNC improves agility by automating compute, storage and networking, reducing deployment time and increasing flexibility.

## ⑤ Identify mismatch in Multi-cloud Federation.

Problem Analysis :- Different cloud providers use different identities

Systems, authentication methods, and user formats. This causes

Login failures and inconsistent access control.

## Secure identity and privacy design:-

- Use Federated Identity Management (FIM).
- Implement Single Sign-On (SSO)
- Use Standard protocols such as SAML 2.0, OAuth 2.0 or OpenID Connect
- Apply Multi-Factor Authentication (MFA)
- Use Role-Based Access Control (RBAC)
- Encrypt Identity data during transmission and storage
- Apply the principle of least privilege.

## Conclusions:-

A federated identity solution using SSO, standard authentication protocols, MFA, and RBAC provides secure and seamless access across multiple cloud providers while protecting user privacy.